Bartosz Włodarczyk

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Education

2024 – 2027 **Delft University of Technology** - Bachelor of Computer Science and Engineering

2020 – 2024 **Batory High School in Warsaw** - International Baccalaureate Diploma Programme (Higher Level: Mathematics Analysis and Approaches, Physics, English)

Skills

Languages
- **English** - fluent
- **German** - B2+
- **Dutch** - communicative
- **Polish** - native

Programming
- Java, SQL, Python, LaTeX, some experience in: C++, Rust

Web Dev
- HTML, CSS, JavaScript, TypeScript, Tailwind CSS, Next.js, Svelte, Angular

Employment History

2025 – present **Teaching Assistant**, **Developer** at PRIME, TU Delft - I am creating educational resources for Mathematics courses offered in engineering programs at the university. Currently, I am developing web-based applets (interactive animations) used in mathematics textbooks and the framework used to create them.

2024 – 2025 **Mathematics Tutor** - I have provided professional Mathematics tutoring to high school students, in both English and Polish. I also worked with tutoring agencies such as *Acady*.

Awards and Achievements

2024 **Best mathematician graduate award**, *Łukasiewicz award*, awarded by my high school annually to the best graduate in the field of Mathematics.

2023 **Honorable mention**, 9th annual International Mathematical Modelling Challenge, a contest organised, among others, by Freudenthal Institute.

Special recognition award, *Matużytkownicy* national mathematical modelling contest organised by Warsaw University of Technology.

2022 **Merit Award**, Singapore International Mathematical and Computational Senior Challenge organised by National University of Singapore High School - won the silver medal.

Top 10% rank in *Náboj* 2022 Juniors international mathematical competition organised, among others, by ETH Zürich and University of Cambridge.

Experience

Training courses

2023 Series of robotics and **Arduino** workshops at Warsaw University of Technology

2019 **Python** "Young Programme" programming course at Warsaw University of Technology

2018 **C++** "Young Programmer" programming course at Warsaw University of Technology

Professional experience

2024 – 2025 **NetNote** - a team project I created for my CSE course at university. The program is a note taking application with various customizations (e.g., themes, tags, and note collections) and server syncing. It was made in Java using JavaFX and Spring.

2024 **QuickQueue** - a demo of a queue system with numbered tickets written in Next.js and using PostgreSQL for database. Source code: `https://github.com/bewuwy/quick-queue`

2023 **Battlecode 2023** - I worked together in a group to program an autonomous bot (in Java) for the tournament organised by MIT. Source code: `https://github.com/bewuwy/bc-23`

Neo Anki Leaderboard - a simple extension for the Anki program adding a leaderboard written in Python. I also created a website in SvelteKit using a PostgreSQL database to host the leaderboard online. Source code: `https://github.com/bewuwy/Neo-Anki-Leaderboard`

Blog website - I designed and deployed a simple blog site using 11ty and Tailwind CSS. Source code: `https://github.com/bewuwy/the-blog`

2022 – 2023 **revers.io** - an online implementation of the Reversi board game supporting multiplayer games. The website was designed in Figma, written in Angular, and uses Google's Firebase database. Source code: `https://github.com/bewuwy/revers.io/`

2022 **Airplane boarding simulator** - I created this simulator in Python using PyGame for a math modeling problem I had to solve. Source code: `https://github.com/bewuwy/airplane-boarding`

Flappy Bevy - a simple clone of Flappy Bird written in Rust programming language using the Bevy game engine. Source code: `https://github.com/bewuwy/flappy_bevy`

2020 – 2024 **Personal website** - my personal (`bewu.dev`) website, which I have been designing and changing for the past four years. Currently running on Next.js with a custom CSS stylesheet. Source code: `https://github.com/bewuwy/bewuwy`

Academic experience

2023 – 2024 **Mathematics Extended Essay** - "Exploring mathematics behind the Cody Dock bridge". I wrote a 30-page analysis of an unusual bridge located in London and how catenary curves were used in its construction. The topic of my research was calculating the road that the bridge "rolled on" using calculus and Python-based approximations. This work was part of my International Baccalaureate diploma. Link to the essay: `https://bewu.dev/ee-pdf`

2023 **Warsaw Science Slam** (a science conference) - My presentation "The quietest theft of the century - How AI stole art." was about the use of AI in art and a dive into how the image generating models like Stable Diffusion work.